Rifat Rafiu

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Academic credentials

Bachelor of Science in Materials Science and Engineering ***March 2022–Present***
Khulna University of Engineering & Technology (KUET)
CGPA: **3.55**/4.00 (up to 4th Year, 1st Semester)

Research Interest

Nanomaterials, Nanoparticle synthesis, Sustainable energy technologies, Optoelectronic properties and applications, Computational materials science, Density functional theory.

Research Experience

Undergraduate Research Work ***September 2025– Present***
Khulna University of Engineering & Technology (KUET)
Photocatalytic Performance of Cobalt Oxide Nanostructures: A Combined Experimental-DFT Investigation

- Synthesized and characterized Co_3O_4 nanostructures using a co-precipitation method for photocatalytic applications.
- Investigated photocatalytic degradation performance by correlating optical band gap, charge transport, and surface properties.
- Applied DFT calculations alongside experiments to analyze structural and charge morphology, optical, and photocatalytic activity enhancement.

Publications

1. **Rafiu, R.,** Hasan Saikot, M. S., Apon, I. A., Boukhris, I., El-Rayyes, A., Khan, M. T., ... & Rahman, M. A. (2025). Comprehensive DFT and SCAPS-1D Study of Structural, Electronic, Optical, Mechanical, Phonon, Thermoelectric, and Photovoltaic Properties of Lead-Free Z3BrO (Z= K, Rb, Cs, and Fr) Anti-Perovskites. Journal of Computational Chemistry, 46(24), https://doi.org/10.1002/jcc.70221
2. **Rafiu, R**., Kriaa, K., Saikot, M. S. H., Rahman, M. A., Apon, I. A., Sfina, N., ... & Talukder, I. A. (2025). Comprehensive first-principles analysis and device simulations of vacancy-ordered D2CeX6 double perovskites for high-efficiency lead-free solar cells. *RSC advances*, *15*(52), 44711-44748. https://doi.org/10.1039/d5ra07162c
3. **Rafiu, R.**, Hasan, S., El-Rayyes, A., Apon, I. A., Shkir, M., Khan, M. T., ... & Rahman, M. A. (2025). Exploring A2CeCl6 (A= K, Rb, Cs, Fr) lead-free double perovskites via first-principles and device simulation for photovoltaic applications. Journal of Physics and Chemistry of Solids, 113126. https://doi.org/10.1016/j.jpcs.2025.113126
4. **Rafiu, R.,** Hasan, M. S., Rahman, M. A., Apon, I. A., Kriaa, K., Benghanem, M., ... & Elboughdiri, N. (2026). First-principles calculations to investigate structural, electronic, optical, elastic, mechanical and phonon properties of novel Q3GaBr6 (Q= Na and K) for next-generation lead-free solar cells. *RSC Advances*, *16*(9), 7803-7829. https://doi.org/10.1039/D5RA10011A
5. **Rafiu, R.,** Apon, I. A., Benghanem, M., Hasan, M. R., Ovi, I. A., Ashraf, I. M., ... & Islam, M. M. (2026). Pressure-Tuned Li-Based Perovskites: A DFT-Guided Device-Level Design for Next-Generation Solar Cells. *Optics Communications*, 133084. https://doi.org/10.1016/j.optcom.2026.133084
6. **Rafiu, R**., Rahman, M. A., Hasan, M. S., Apon, I. A., Kashif, M., Ramalingam, R. J., ... & Al-Haidar, F. (2026). DFT-based study on the physical and molecular dynamics properties of some E3IO alkaline-metal anti-perovskites for renewable energy applications. *International Journal of Modern Physics B*, *40*(08), 2650078. https://doi.org/10.1142/S0217979226500785

7. **Rafiu, R**., Kriaa, K., Rahman, M. A., Apon, I. A., AlFaify, S., Maatki, C., ... & Elboughdiri, N. (2026). Unveiling the stability and optoelectronic potential of alkali-based L3FO anti-perovskites: a first-principles perspective. The European Physical Journal Plus, 141(1), 32. https://doi.org/10.1140/epjp/s13360-025-07239-3 .
8. Mahmud, M. N., **Rafiu, R**., Paul, S. K., Enam, S. E., & Hasan, M. M. (2026, January). Hydrothermally Synthesized Urea-Derived Co3O4 Nanoparticles for Efficient Photocatalytic Degradation of Organic Dye. In *2026 5th International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE)* (pp. 1-6). IEEE. https://doi.org/10.1109/ICECTE69292.2026.11429372
9. Rahman, M. A., **Rafiu, R.,** Apon, I. A., Saikot, M. S. H., Ovi, I. A., Elboughdiri, N., ... & Albalawi, H. (2026). Pressure-tunable optoelectronic and photovoltaic properties of lead-free ZnAgF3 and ZnAg(F0.5Cl0.5)3 Perovskites: A first-principles and device-level study. Materials Chemistry and Physics, 349, 131777. https://doi.org/10.1016/j.matchemphys.2025.131777
10. Saikot, M. S. H., **Rafiu, R**., Apon, I. A., El-Rayyes, A., Rahman, M. A., Shkir, M., ... & Marnadu, R. (2025). First-principles investigation of structural, electronic, optical, mechanical, and phonon properties of Pb-and Sn-based cubic oxide perovskites for optoelectronic applications. RSC advances, 15(46), 39071-39091. 10.1039/D5RA06065F .

Technical skills

- **Simulation Software** — CASTEP, VESTA, CES Edupack, LAMMPS, SCAPS-1D
- **Computer-aided Design** — AutoCAD
- **Word Processing and Presentation** — MS Word, MS PowerPoint
- **Graphing and Data Analysis** — Origin Pro, MS Excel
- **Programming** — Python, C, C^{+}
- **Metallurgical Analysis** — Metallurgical Microscope, XRD, FTIR, SEM
- **Synthesis** — Hydrothermal synthesis, Co-precipitation method
- **Hands-on** — Universal Testing Machine, Brinell and Rockwell hardness test, Manufacturing shop equipment (Welding, Machining, Milling), Furnace operation.
- **Linguistic Proficiency** — Bangla (Native), English, Hindi

Academic Awards and Honors

- **Government Scholarships:** Awarded by the Government for achieving outstanding academic results in Junior School Certificate (JSC), Secondary School Certificate (SSC), and Higher Secondary Certificate (HSC) examinations.

Training

- **BSRM Steels Ltd.,** Chattogram, Bangladesh — ***27th February, 2025 – 5th March,2025***

Co-Curricular Activities

- **Member of Cultural Committee** — ***October 2024 – October 2025***

  Amar Ekushey Hall, Khulna University of Engineering and Technology, Khulna

  Planned, coordinated, and executed cultural programs, enhancing student engagement through effective teamwork, communication, and organizational management.
- **QnA Teacher**

  Udvash Academic and Admission Care — ***March 2023 – August 2025***

  Conducted Q&A-based academic sessions, clarifying concepts and guiding students to improve problem-solving skills and exam performance.